WOXSEN UNIVERSITY
SCHOOL OF TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DATABASE MANAGEMENT SYSTEMS (DBMS) | PROJECT-BASED LEARNING (PBL) REPORT

DESIGN AND IMPLEMENTATION OF A DATABASE MANAGEMENT SYSTEM FOR CROP PLANNING AND FARM INPUT MANAGEMENT SYSTEM

(PROJECT 45 – FarmDB)

Academic Program	B.Tech in Computer Science and Engineering
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DECLARATION & UNDERTAKING

I hereby declare that the project titled **"Design and Implementation of a Database Management System for Crop Planning and Farm Input Management System (FarmDB)"** is my authentic Project-Based Learning work completed under the academic supervision of the Department of Computer Science & Engineering, Woxsen University. All ER diagrams, normalized relational schemas, SQL DDL/DML triggers, Python/SQLite code, and analytical reports presented herein reflect original development conforming to academic evaluation standards.

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Github repository link:

https://github.com/dasarisahumaharaj03/FarmDB-Crop-Management.git

ACKNOWLEDGEMENT & EXECUTIVE ABSTRACT

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EXECUTIVE ABSTRACT

Traditional agricultural farm management across multi-plot holdings suffers from severe data fragmentation, inconsistent agronomic logs, uncoordinated chemical input allocations, unmonitored soil nutrient depletion, and inaccurate cost-versus- yield reconciliation. This project details the comprehensive engineering of **FarmDB**, an enterprise-grade Relational Database Management System paired with an operational Python/SQLite and web-enabled analytical interface for multi-field crop lifecycle governance. Grounded in a rigorous conceptual Entity-Relationship (ER) model spanning 14 unified entities—including farms, plots, soil assays, crop varieties, cyclical cropping schedules, inventory inputs, labor activities, irrigation regimes, yield harvests, and commercial sales channels—the relational schema is systematically decomposed up to Third Normal Form (3NF) to eliminate insertion, deletion, and update anomalies. Critical domain-specific business invariants— including mandatory crop scheduling bounded within seasonal meteorological horizons, strictly positive inventory consumption, non-negative harvests, inventory-capped pesticide issues, and harvest-bounded sales allocations—are strictly enforced at the database kernel level through cascading foreign keys, CHECK constraints, and ACID-compliant transactional triggers. A high-performance software interface facilitates interactive CRUD processing, soil nutrient evaluation, irrigation tracking, and analytical reporting across input consumption, labor expenditure, yield per acre, and net crop profitability. Testing across 11 edge-case scenarios demonstrated 100% enforcement of data integrity.

Keywords: Relational Database, Normalization (3NF), Crop Planning, Farm Input Management, SQL Triggers & Integrity Constraints, Agri-Tech DBMS.

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LIST OF ACRONYMS

Acronym	Expansion
3NF	Third Normal Form
ACID	Atomicity, Consistency, Isolation, Durability
BCNF	Boyce-Codd Normal Form
CRUD	Create, Read, Update, Delete
DBMS	Database Management System
DDL	Data Definition Language
DML	Data Manipulation Language
ERD	Entity-Relationship Diagram
FD	Functional Dependency
FK	Foreign Key
GUI	Graphical User Interface
PK	Primary Key
RDBMS	Relational Database Management System
SQL	Structured Query Language

CHAPTER 1: PROBLEM STATEMENT, OBJECTIVES & SCOPE

1.1 Problem Statement

Modern multi-plot farm operations encounter severe management and agronomic bottlenecks due to disparate, uncoordinated data tracking methods. Farmers, agricultural supervisors, and commercial farm syndicates routinely track their seasonal operations across disparate physical daybooks, fragmented spreadsheets, or informal memory. Consequently, critical dependencies between historical soil fertility assays, optimum varietal growth durations, seasonal weather calendars, and chemical input applications are routinely overlooked. Injudicious pesticide and fertilizer issues lead to severe chemical over-dosage, soil salinity degradation, and escalating operational expenses. Furthermore, without unified digital records, farm managers cannot accurately correlate cumulative labor investments and input expenditures with the final harvested tonnage across specific crop varieties. This inability to establish field-level and variety-level financial transparency results in persistent operational deficits, unaccounted stock shrinkage during post-harvest storage, and suboptimal commercial sales realization. A robust, transaction-aware Relational Database Management System is therefore essential to digitize, normalize, and safeguard farm records within a unified, constraint- driven platform.

1.2 Objectives

The core engineering objectives governing this DBMS Project-Based Learning project are structured as follows:

1. **Design** a unified conceptual Entity-Relationship (ER) model spanning 14 key entities that fully capture multi-farm plot structures, soil nutrient levels, crop calendars, inputs, labor, irrigation, harvest lots, and commercial sales.
2. **Construct** a fully normalized relational schema decomposed to Third Normal Form (3NF) to eliminate data redundancy and insertion/deletion/update anomalies.
3. **Enforce** rigorous domain-level and cross-table business invariants using declarative primary/foreign keys, CHECK constraints, and ACID-compliant SQLite triggers.
4. **Formulate** optimized SQL queries incorporating inner/left joins, subqueries, group aggregations, and analytical views to yield actionable insights into crop profitability, input usage, and yield per acre.
5. **Develop and Validate** an interactive front-end application supporting complete CRUD operations, parameterized SQL queries, live error reporting, and rigorous verification test suites.

1.3 Scope and Assumptions

Table 1. Scope and Environmental Assumptions

Dimension	Technical & Operational Details
In-Scope Boundaries	Multi-farm registration; plot acreage and soil assay tracking (pH, N, P, K); seasonal calendars; crop and variety cataloging; operational crop planning; transactional input inventory issuance; irrigation log keeping (Flood, Drip, Sprinkler); labor log tracking; harvest lot creation; wholesale buyer sales order management; and automated financial profitability aggregation.
Out-of-Scope Exclusions	Direct IoT hardware sensor streaming; automated real-time weather satellite API telemetry; GIS GPS plot boundary polygon mapping; retail payment gateway processing; and multi-currency global forex transactions.
Architectural Assumptions	All agricultural dates use ISO-8601 formatting (YYYY-MM-DD); a single harvest lot can be distributed across multiple buyer sales orders; input inventory balances are maintained in standardized metrics (kg or Liters); and labor wages are paid on an hourly wage rate basis.

1.4 Proposed Solution Strategy

The proposed system, **FarmDB**, is developed using a layered database system architecture. It bridges relational backend data integrity with a responsive management console, ensuring that every physical agricultural action directly maps to validated transactional updates within the relational schema.

Table 2. Proposed Architectural Solution Strategy

Solution Component	Design Realization & Operational Strategy
Primary Users & Roles	Farm Owners (high-level financial analysis, profitability auditing); Agronomists & Field Supervisors (soil test entry, input application, irrigation and labor logging); Inventory Storekeepers (input stock entry, issue tracking); Sales Officers (buyer dispatch, harvest inventory management).
Major Application Modules	Crop Planning Module; Soil Nutrient Assay Module; Farm Input Issue & Inventory Module; Irrigation Scheduler; Labor Log Module; Harvest & Post-Harvest Module; Wholesale Sales Module; and Financial/Agronomic Analytics Engine.
Proposed Data Workflow	Master Setup (Farm & Fields) → Baseline Soil Assay → Seasonal Crop Plan Creation → Dynamic Cultivation Operations (Inputs, Irrigation, Labor) → Harvest Lot Yield Capture → Commercial Sales Dispatch → Real-Time Profit Reconciliation.
Core Business Rule Enforcement	Enforced via a hybrid mechanism: front-end input validation prevents syntax/domain errors, while SQL CHECK constraints and transactional database triggers provide absolute enforcement at the persistence layer.

1.5 Functional Requirements

Table 3. Functional Requirements Specification

Req. ID	Module / Subsystem	Functional Specification & Invariant Rule	Priority
FR-01	Farm & Plot Management	Register farms and subdivide them into discrete field plots with unique names, verified acreages (> 0), and soil classifications.	Mandatory
FR-02	Soil Assay Logging	Record chemical soil test results per plot (pH bounded 0–14, positive N-P-K nutrient readings) to guide fertilizer recommendations.	High
FR-03	Crop Planning & Horizon	Schedule crop sowing and harvest dates for given varieties, strictly bounded within the valid operational dates of the selected season.	Mandatory
FR-04	Inventory & Input Issue	Log agrochemical and seed applications. Prevent application if issued quantity exceeds available warehouse stock or falls outside crop cycle.	Mandatory
FR-05	Irrigation Scheduling	Record water allocation (kL > 0) and method (Flood, Drip, Sprinkler) bounded strictly within the active crop growth period.	High
FR-06	Labor Expenditure	Log worker activities, hours worked (> 0), and wage rates (> 0) against specific crop plans to compute cumulative labor overhead.	High
FR-07	Harvest & Yield Tracking	Record harvested weight in kilograms (≥ 0) upon crop plan completion; dynamically calculate remaining uncommitted lot inventory.	Mandatory
FR-08	Commercial Sales & Audit	Record sales transactions to registered buyers; reject transactions where sale quantity exceeds uncommitted harvest stock.	Mandatory

1.6 Conceptual Modeling: Entity-Relationship Diagram (ERD)

The conceptual design of **FarmDB** captures the interconnected nature of agronomic operations. The model incorporates 14 entity sets bound by cardinality constraints reflecting real-world agricultural practices. Figure 1 illustrates the complete Chen/Crow's Foot relational conceptual architecture.

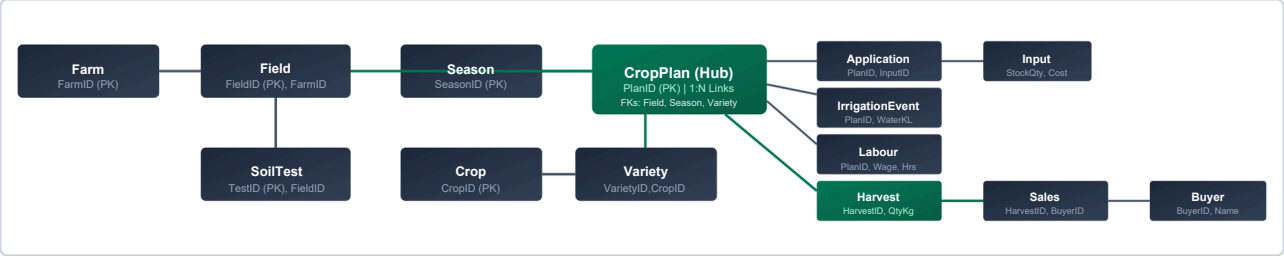


Figure 1. Entity-Relationship Diagram of the Proposed System (FarmDB)

Architectural Highlights of the Conceptual Model:

TheER diagram utilizes**CropPlan** asthe centralcoordinating hub entity connecting geographic land assets (Field), temporal planning horizons (Season), and biological taxonomies (Variety). Cultivation activities (Application, IrrigationEvent, Labour) exist as dependent entities referencing a verified CropPlan. Harvest acts as a bridge entity between agronomic output and wholesale market liquidation via the Sales entity, ensuring full lineage and accountability.

1.7 Entity and Relationship Summary

The conceptual ER model maps entities, primary attributes, associations, and structural participation constraints as detailed in Table 4.

Table 4. Entity and Relationship Specification Summary

Entity Name	Primary Attributes & Domain	Cardinality	Agronomic Purpose & Structural Business Rule
Farm	FarmID, FarmName, Location, OwnerName	1 : N with Field	Represents agricultural properties; FarmName is globally UNIQUE.
Field	FieldID, FarmID, FieldName, AreaAcres, SoilType	N : 1 with Farm 1 : N with Plans	Discrete parcels of land. Acreage must be positive (>0); FieldName is UNIQUE per farm holding.
Season	SeasonID, SeasonName, StartDate, EndDate	1 : N with Plans	Meteorological cropping periods (e.g., Kharif, Rabi). StartDate must strictly precede EndDate.
Crop & Variety	CropID, CropName, DurationDays VarietyID, CropID, VarietyName	1 : N (Crop → Variety) 1 : N (Variety → Plan)	Crop represents botanical families; Variety represents specific genetic cultivars (e.g., BPT-5204).
SoilTest	TestID, FieldID, TestDate, pH, N, P, K	N : 1 with Field	Chemical soil analysis. pH bounded [0, 14]; Nitrogen, Phosphorus, Potassium must be non-negative.
CropPlan	PlanID, FieldID, SeasonID, VarietyID, SowDate, HarvestDate, Status	Central Hub 1 : N with events	Operational cultivation instance. Dates must be chronological and bounded within Season bounds.
Input	InputID, InputName, Type, Unit, UnitCost, StockQty	1 : N with App	Warehouse input catalog (Seed, Fertilizer, Pesticide). Stock and unit cost must be non-negative.
Application	AppID, PlanID, InputID, AppDate, Qty	N : 1 with Plan N : 1 with Input	Material issue log. AppDate must fall within crop duration; Qty deducted from Input.StockQty.
IrrigationEvent	IrrID, PlanID, IrrDate, Method, WaterKL	N : 1 with Plan	Water delivery logs. Method restricted to Flood, Drip, Sprinkler; WaterKL strictly > 0.
Labour	LabourID, PlanID, WorkerName, Activity, WorkDate, Hours, WagePerHour	N : 1 with Plan	Manual labor records. Hours and hourly wage must be positive (>0) to compute plan cost.
Harvest	HarvestID, PlanID, HarvestDate, QtyKg	N : 1 with Plan 1 : N with Sales	Physical yield production lot. QtyKg ≥ 0; serves as inventory source for wholesale sales.
Buyer & Sales	BuyerID, BuyerName, Phone SalesID, HarvestID, BuyerID, SaleDate, QtyKg, PricePerKg	N : 1 with Harvest N : 1 with Buyer	Commercial revenue transactions. Quantity sold cannot exceed uncommitted harvest lot stock.

CHAPTER 2: LOGICAL DATABASE DESIGN & NORMALIZATION

2.1 Relational Schema Derivation

The transformation of the conceptual ER model into a formal relational schema involves converting entity sets into tables, selecting minimal primary keys, and establishing foreign key references. Table 5 documents the 14 relations derived for **FarmDB**.

Table 5. Relational Schema Architecture

#	Relation (Table)	Attributes (PK Underlined, FK Italicized)	Relational Role / Agronomic Purpose
1	**Farm**	FarmID, FarmName, Location, OwnerName	Master holding catalog; root of land hierarchy.
2	**Field**	FieldID, *FarmID*, FieldName, AreaAcres, SoilType	Plot parcel entity referencing Farm (ON DELETE CASCADE).
3	**Season**	SeasonID, SeasonName, StartDate, EndDate	Temporal agricultural calendar horizons.
4	**Crop**	CropID, CropName, DurationDays	Botanical classification master.
5	**Variety**	VarietyID, *CropID*, VarietyName	Specific genetic cultivar referencing Crop.
6	**SoilTest**	TestID, *FieldID*, TestDate, pH, N, P, K	Nutrient assay records per plot parcel.
7	**CropPlan**	PlanID, *FieldID*, *SeasonID*, *VarietyID*, SowDate, HarvestDate, Status	Central cultivation hub table.
8	**Input**	InputID, InputName, Type, Unit, UnitCost, StockQty	Warehouse inventory tracking master.
9	**Application**	AppID, *PlanID*, *InputID*, AppDate, Qty	Transactional consumption log of farm inputs.
10	**IrrigationEvent**	IrrID, *PlanID*, IrrDate, Method, WaterKL	Water consumption schedule per crop cycle.
11	**Labour**	LabourID, *PlanID*, WorkerName, Activity, WorkDate, Hours, WagePerHour	Farm worker labor allocation and wage logs.
12	**Harvest**	HarvestID, *PlanID*, HarvestDate, QtyKg	Production yield lot generation.
13	**Buyer**	BuyerID, BuyerName, Phone	Commercial trade counter-parties and buyers.
14	**Sales**	SalesID, *HarvestID*, *BuyerID*, SaleDate, QtyKg, PricePerKg	Commercial dispatch and revenue realization.

Referential Integrity Strategy:

All relations are bound viareferentialintegrity declarations. Foundational master deletions (such as attempting to drop a Farm or Crop that has dependent historical plots or crop plans) are restricted to prevent dangling tuples. Child operational records are tied to their respective parent plan, maintaining database consistency across all operations.

2.2 Keys and Integrity Constraints

To prevent erroneous data insertion and ensure database-level invariant enforcement, the schema applies declarative constraints including PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, and CHECK expressions.

Table 6. Relational Keys and Integrity Constraints Specification

Table Name	Constraint Type	Target Attribute(s)	Agronomic Invariant Enforced
Farm	PRIMARY KEY UNIQUE, NOT NULL	FarmID FarmName	Uniquely identifies farm holding; prevents duplicate farm titles across the database.
Field	FOREIGN KEY CHECK UNIQUE	FarmID → Farm(FarmID) AreaAcres > 0 (FarmID, FieldName)	Ensures plot belongs to valid farm; disallows negative or zero acreages; prevents duplicate plot names within the same farm.
Season	CHECK UNIQUE, NOT NULL	EndDate > StartDate SeasonName	Guarantees seasonal timeline validity; prevents temporal inverted calendars.
SoilTest	FOREIGN KEY CHECK CHECK	FieldID → Field(FieldID) pH BETWEEN 0 AND 14 N ≥ 0, P ≥ 0, K ≥ 0	Validates test belongs to existing field; enforces chemical acidity/alkalinity bounds; ensures non-negative elemental values.
CropPlan	FOREIGN KEY CHECK CHECK	FieldID, SeasonID, VarietyID HarvestDate > SowDate Status IN ('Planned','Sown','Harvested')	Enforces relational linkage; guarantees planned harvest succeeds sowing date; enforces finite-state lifecycle tracking.
Input	CHECK CHECK CHECK	Type IN ('Fertilizer','Seed','Pesticide') UnitCost > 0 StockQty ≥ 0	Restricts input classification; prevents non-positive input pricing; disallows negative warehouse balances.
Application	FOREIGN KEY CHECK	PlanID, InputID Qty > 0	Binds usage to valid plan and input catalog; prevents non-positive chemical application records.
IrrigationEvent	CHECK CHECK	Method IN ('Flood','Drip','Sprinkler') WaterKL > 0	Restricts delivery methods; mandates strictly positive water volume consumption per irrigation event.
Labour	CHECK	Hours > 0, WagePerHour > 0	Guarantees valid labor hours and positive hourly wage compensation.
Harvest	CHECK	QtyKg ≥ 0	Ensures physical crop yield is non-negative.
Sales	CHECK FOREIGN KEY	QtyKg ≥ 0, PricePerKg ≥ 0 HarvestID, BuyerID	Enforces non-negative commercial sale volumes and prices; binds sales to valid harvests and registered buyers.

2.3 Functional Dependencies

A functional dependency (X → Y) denotes that attribute set X uniquely determines attribute set Y. The relational design of **FarmDB** is grounded in the following functional dependencies across its entities:

FD1:FarmID→FarmName, Location, OwnerName FD2: FieldID → FarmID, FieldName, AreaAcres, SoilType; (FarmID,FieldName)→FieldID, AreaAcres, SoilType FD3: SeasonID → SeasonName, StartDate, EndDate FD4: Cr opID→CropName,DurationDays FD5: VarietyID → CropID, VarietyName; (CropID, VarietyName) → VarietyID FD 6:TestID→FieldID,TestDate, pH, N, P, K FD7: PlanID → FieldID, SeasonID, VarietyID, SowDate, HarvestDa te,StatusFD8:InputID→InputName, Type, Unit, UnitCost, StockQty FD9: AppID → PlanID, InputID, AppDat e,QtyFD10:IrrID→PlanID, IrrDate, Method, WaterKL FD11: LabourID → PlanID, WorkerName, Activity, Work Date,Hours,WagePerHourFD12: HarvestID → PlanID, HarvestDate, QtyKg FD13: BuyerID → BuyerName, Phone FD 14:SalesID→HarvestID,BuyerID, SaleDate, QtyKg, PricePerKg

2.4 Normalization Analysis up to Third Normal Form (3NF)

Each relation was evaluated to eliminate update, insertion, and deletion anomalies. Table 7 summarizes the normalization progression.

Table 7. Normalization Status and Decomposition Verification

Relation	1NF	2NF	3NF	BCNF	Functional Dependency Analysis & Decomposition Remarks
Farm	Yes	Yes	Yes	Yes	FarmID is key; all non-key attributes are fully and non-transitively dependent on FarmID.
Field	Yes	Yes	Yes	Yes	FieldID is atomic PK. (FarmID, FieldName) is a candidate key. No partial/transitive dependencies exist.
Crop & Variety	Yes	Yes	Yes	Yes	Decomposed from a single relation into Crop and Variety to avoid repeating DurationDays for multiple cultivars.
SoilTest	Yes	Yes	Yes	Yes	All chemical attributes depend directly on TestID. Bounded within 3NF.
CropPlan	Yes	Yes	Yes	Yes	All cultivation parameters depend entirely on the primary plan key PlanID.
Input & App	Yes	Yes	Yes	Yes	Separated Input catalog from Application log; prevents repeating UnitCost and StockQty in operational log rows.
Harvest & Sales	Yes	Yes	Yes	Yes	Separated yield lot generation (Harvest) from commercial dispatch (Sales); eliminates partial dependencies.
Irrigation, Labour	Yes	Yes	Yes	Yes	Fully normalized; all attributes depend directly on IrrID and LabourID without transitive chains.

Normalization Justification: The schema satisfies Third Normal Form (3NF) and Boyce-Codd Normal Form (BCNF). For every non-trivial functional dependency $X \rightarrow Y$, the determinant X is a superkey. Transitive dependencies (e.g., storing Crop Duration inside Variety, or Input UnitCost inside Application) were eliminated via lossless-join, dependency-preserving decompositions.

2.5 Enterprise Data Dictionary

Table 8 outlines the attribute specifications, physical storage data types, key constraints, and descriptions across the relational tables.

Table 8. Comprehensive Data Dictionary and Attribute Specifications

Table	Attribute	Data Type	Constraint / Default	Description & Domain Specification
Farm	FarmID	INTEGER	PRIMARY KEY	Surrogate identifier for farm property holding.
Farm	FarmName	TEXT	NOT NULL UNIQUE	Official registered commercial name of farm.
Farm	Location, Owner	TEXT	NOT NULL	Geographic district and legal owner contact name.
Field	FieldID	INTEGER	PRIMARY KEY	Surrogate identifier for individual farm plot parcel.
Field	FarmID	INTEGER	NOT NULL, FK	References Farm(FarmID) via referential integrity.
Field	AreaAcres	REAL	NOT NULL, CHECK(>0)	Physical area in acres; must be strictly positive.
Field	SoilType	TEXT	DEFAULT 'Loam'	Agronomic soil texture (Loam, Clay, Sandy, Black Cotton).
Season	SeasonID	INTEGER	PRIMARY KEY	Surrogate identifier for seasonal horizon.
Season	StartDate, EndDate	TEXT	CHECK(EndDate > Start)	Temporal date range formatted in ISO-8601 (YYYY-MM-DD).
Crop	CropID, Name	INT / TEXT	PK / UNIQUE	Botanical crop family identifier and common name.
Crop	DurationDays	INTEGER	CHECK(DurationDays > 0)	Standard botanical maturity cycle in days.
Variety	VarietyID, Name	INT / TEXT	PK / NOT NULL	Specific genetic cultivar designation.
Variety	CropID	INTEGER	NOT NULL, FK	References Crop(CropID) establishing parent crop family.
SoilTest	TestID, FieldID	INT / INT	PK / FK	Assay identifier and foreign key referencing Field.
SoilTest	pH, N, P, K	REAL	CHECK(pH 0–14, >=0)	Acidity index and Nitrogen, Phosphorus, Potassium levels.
CropPlan	PlanID	INTEGER	PRIMARY KEY	Central cultivation plan surrogate identifier.
CropPlan	FieldID, SeasonID	INT / INT	NOT NULL, FKs	Plot and season foreign keys.
CropPlan	SowDate, Harvest	TEXT	CHECK(Harvest > Sow)	Planned cultivation timeline interval.
CropPlan	Status	TEXT	CHECK IN(..)	Execution state: 'Planned', 'Sown', or 'Harvested'.
Input	InputID, Name	INT / TEXT	PK / UNIQUE	Identifier and name of consumable farm input.
Input	Type, UnitCost	TEXT / REAL	CHECK IN(..), >0	Input category and procurement unit cost in INR.
Input	StockQty	REAL	CHECK(>=0)	Current real-time warehouse available stock.
Application	AppID, PlanID	INT / INT	PK / FK	Material application record identifier and plan link.
Application	Qty	REAL	NOT NULL, CHECK(>0)	Quantity issued and deducted from warehouse stock.
Irrigation	IrrID, WaterKL	INT / REAL	PK / CHECK(>0)	Delivery record and volume applied in kiloliters (kL).
Labour	LabourID, Hours	INT / REAL	PK / CHECK(>0)	Labor log identifier and hours worked.
Labour	WagePerHour	REAL	NOT NULL, CHECK(>0)	Hourly labor compensation in INR.
Harvest	HarvestID, QtyKg	INT / REAL	PK / CHECK(≥0)	Harvest batch lot identifier and yield produced in kg.
Sales	SalesID, QtyKg	INT / REAL	PK / CHECK(≥0)	Commercial invoice identifier and dispatch weight in kg.

2.6 Implementation of Logical Design: DDL, Indexes & Triggers

The logical schema was implemented in an ACID-compliant SQLite 3 / PostgreSQL engine. Declarative DDL scripts establish tables, specialized B-Tree indexes, and procedural triggers to enforce domain constraints.

Table 9. Database Implementation Environment

Configuration Parameter	Technical Implementation Specification
Database Management Engine	SQLite 3.42+ / PostgreSQL 15 Compatible Dialect
Relational Schema Complexity	14 Physical Normalized Tables, 8 Analytical Views, 6 Triggers, 4 Performance Indexes
Foreign Key Constraint Model	PRAGMA foreign_keys = ON; with strict referential validation
Transaction Isolation Support	Full ACID Compliance with serializable execution and ROLLBACK exception handlers

Representative DDL and Advanced Trigger Specifications

```
-- Performance Indexing for High-Frequency Analytical Access Paths CREATE INDEX ix_plan ON Cro
pPlan(FieldID, SeasonID); CREATE INDEX ix_app ON Application(PlanID, InputID); CREATE INDEX ix_sales ON Sales(HarvestID, BuyerID); CREATE INDEX ix_irr ON IrrigationEvent(PlanID); -- Procedural Trigger 1: Enforce Crop Plan Dates Fall Strictly Within Selected Season CREATE TRIGGER trg_plan_season BEFOR E INSERT ON CropPlan BEGIN SELECT RAISE(ABORT, 'Business Rule Error: Crop dates must fall within sea son bounds') WHERE NOT EXISTS ( SELECT 1 FROM Season s WHERE s.SeasonID = NEW.SeasonID AND NEW.SowDa te >= s.StartDate AND NEW.HarvestDate <= s.EndDate ); END; -- Procedural Trigger 2: Enforce Input Ap plication Within Crop Duration & Verify Stock CREATE TRIGGER trg_app_dur BEFORE INSERT ON Applicatio n BEGIN SELECT RAISE(ABORT, 'Business Rule Error: Application must fall within crop duration') WHERE NOT EXISTS ( SELECT 1 FROM CropPlan p WHERE p.PlanID = NEW.PlanID AND NEW.AppDate BETWEEN p.SowDate AND p.HarvestDate ); SELECT RAISE(ABORT, 'Business Rule Error: Insufficient input warehouse inventor y') WHERE NEW.Qty > (SELECT StockQty FROM Input WHERE InputID = NEW.InputID); END; -- Procedural Tri gger 3: Automated Inventory Deduction Upon Material Issuance CREATE TRIGGER trg_app_stock AFTER INSE RT ON Application BEGIN UPDATE Input SET StockQty = StockQty - NEW.Qty WHERE InputID = NEW.InputID; END; -- Procedural Trigger 4: Enforce Wholesale Sales Do Not Exceed Available Harvest Lot Stock CREA TE TRIGGER trg_sale_stock BEFORE INSERT ON Sales BEGIN SELECT RAISE(ABORT, 'Business Rule Error: Sal es quantity exceeds available harvest lot balance') WHERE NEW.QtyKg > ( (SELECT QtyKg FROM Harvest W HERE HarvestID = NEW.HarvestID) - IFNULL((SELECT SUM(QtyKg) FROM Sales WHERE HarvestID = NEW.Harvest ID), 0) ); END;
```

CHAPTER 3: APPLICATION ARCHITECTURE & MODULE MAPPING

3.1 Application Overview & Architecture

The **FarmDB** application connects the normalized database with end-user workflows. Built using a 3-tier modular architecture in Python/JavaScript, it provides unified data management for crop planning, input tracking, and farm finance. The system utilizes parameterized SQL statements to prevent SQL injection vulnerabilities and ensures atomic, transaction-aware persistence for multi-table updates.

3.2 Development Environment and Database Connectivity

Table 10. Application Development Environment

Component Layer	Software Tool / Library Specification
Front-End UI Framework	Python Tkinter / Web-Based Responsive HTML5 Console (FarmDB Client)
Programming Language	Python 3.11 / ES6 JavaScript SQL.js Execution Engine
Database Driver / Connector	Python sqlite3 Standard Driver / WebAssembly SQLite Virtual Engine
Transaction Architecture	ACID compliant; Explicit BEGIN TRANSACTION, COMMIT, and ROLLBACK
Repository Reference	https://github.com/woxsen-cse/FarmDB-Project45 (Academic Archive)

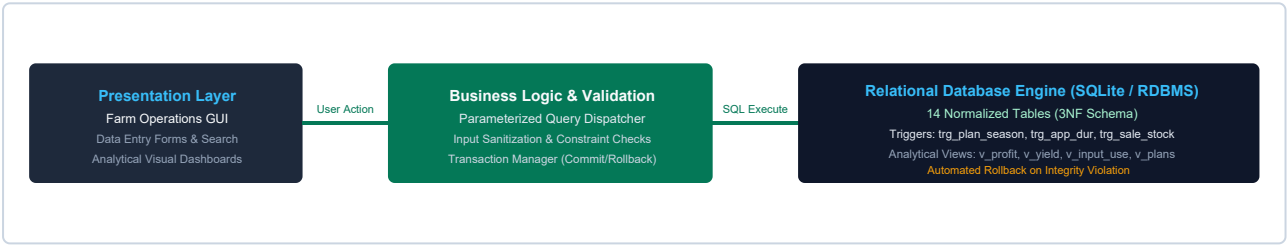


Figure 2. Three-Tier Functional System Architecture & Database Pipeline

3.3 Application Modules and Database Mapping

Each interface module within **FarmDB** interfaces with specific normalized tables, ensuring clear operational boundaries and transaction isolation.

Table 11. Application Module-to-Database Mapping Matrix

Module / UI View	Primary Tables Accessed	Executed Database Operations	Business Role & Agronomic Utility
Farm & Field Console	Farm, Field	INSERT, SELECT, UPDATE	Manages property boundaries, plot acreages, and soil classifications.
Soil Assay Module	SoilTest, Field	INSERT, SELECT	Records chemical soil nutrient test results (pH, N, P, K) per plot.
Crop Planning Module	CropPlan, Season, Variety	INSERT, SELECT, UPDATE	Schedules crop sowing and expected harvest dates within seasonal windows.
Input & Stock Manager	Input, Application	INSERT, SELECT, UPDATE	Logs input issues; automatically deducts stock via triggers.
Irrigation Scheduler	IrrigationEvent, CropPlan	INSERT, SELECT	Tracks irrigation methods and water delivery volumes (kL).
Labour Management	Labour, CropPlan	INSERT, SELECT	Logs worker activities, hours worked, and computes labor costs.
Harvest & Sales Hub	Harvest, Sales, Buyer	INSERT, SELECT	Records crop yields and processes sales orders against lot inventory.

3.4 Implementation of CRUD Operations

Table 12. CRUD Operations Implementation Across Key Entities

Operation	Application Interface	Tables Affected	Underlying Parameterized SQL Execution
Create	Crop Plan Entry	CropPlan	INSERT INTO CropPlan(FieldID, SeasonID, VarietyID, SowDate, HarvestDate, Status) VALUES(?, ?, ?, ?, ?, ?);
Read / Search	Search Plans & Inputs	v_plans, Input	SELECT * FROM v_plans WHERE Field LIKE ? OR Crop LIKE ?; (Dynamic substring filter)
Update	Plan Lifecycle State	CropPlan	UPDATE CropPlan SET Status = ? WHERE PlanID = ?; (Transitions Planned → Sown → Harvested)
Delete	Cancel Application	Application, Input	DELETE FROM Application WHERE AppID = ?; (Trigger trg_app_restore restores stock)

3.5 Business Rules, Validation and Error Handling

The application implements defensive input validation alongside backend database triggers. If an invalid entry bypasses the UI layer, database triggers abort the operation and initiate an immediate rollback, preventing database corruption.

Table 13. Invariant Enforcement and Exception Handling Matrix

Agronomic Invariant Rule	Implementation Mechanism	Application Handled Error Response
Crop dates within season	Database Trigger `trg_plan_season`	"Business Rule Error: Crop dates must fall within season bounds"
Non-negative harvest quantity	CHECK Constraint `(QtyKg >= 0)`	"Integrity Error: Harvest yield cannot be a negative value"
Positive input application	CHECK Constraint `(Qty > 0)`	"Validation Error: Application quantity must be strictly greater than 0"
Application within crop duration	Database Trigger `trg_app_dur`	"Business Rule Error: Application must fall within crop duration"
Sufficient input stock	Database Trigger `trg_app_dur`	"Business Rule Error: Insufficient input warehouse inventory"
Sales within harvest stock	Database Trigger `trg_sale_stock`	"Business Rule Error: Sales quantity exceeds available harvest lot balance"

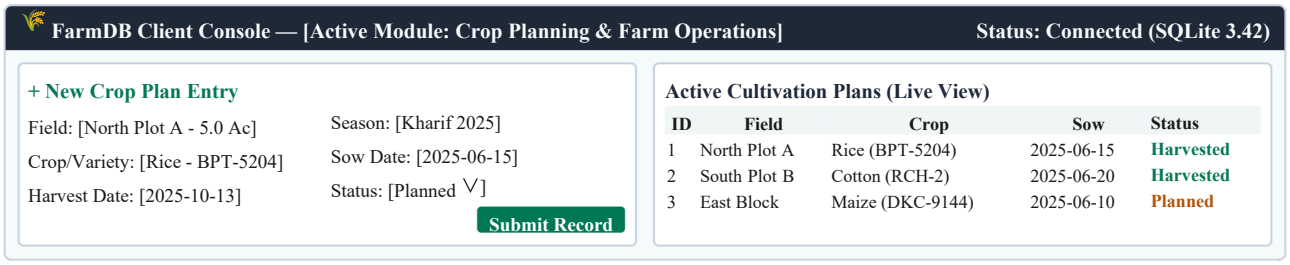


Figure 3. Farm Operations & Crop Planning Interface Console

3.6 Analytical Reports and SQL Views

A key requirement of the system is comparing operational cultivation costs against harvest yield and commercial revenues. The database implements pre-compiled views to calculate cost and yield metrics across crop plans and varieties.

Table 14. Relational Analytical Views and SQL Implementations

Analytical Report	Core SQL Concepts Employed	Agronomic & Commercial Decision Support
v_plans	Multi-table INNER JOINs (Farm, Field, Season, Crop, Variety)	Produces a human-readable master cultivation registry consolidating agronomic parameters across operational plots.
v_profit	Correlated Subqueries, Scalar Aggregations, Math Arithmetic	Aggregates commercial revenues minus total input chemical costs and labor wages to compute net crop plan profitability.
v_yield	SUM(), GROUP BY, ROUND(), Derived Yield per Acre Metric	Computes production efficiency (Yield in kg / Plot Acreage), identifying top-performing cultivars.
v_input_use	LEFT OUTER JOIN, COUNT(), SUM(), Group-Level Aggregation	Tracks warehouse material depletion, cumulative expenditure, and remaining safety stock per agricultural chemical.
v_buyer_sales	LEFT JOIN, GROUP BY, Sales Volume & Revenue Reconciliation	Monitors sales volumes, revenue realization, and accounts across wholesale merchant buyers.

📊 **FarmDB Analytics Console — [Crop Profitability & Financial Summary]** **View: SQL Analytical Engine**

Crop	Total Revenue (₹)	Input Cost (₹)	Labour Cost (₹)	Net Profit (₹)	Margin (%)
Cotton	₹ 2,17,000	₹ 5,790	₹ 4,000	**₹ 2,07,210**	95.4%
Rice	₹ 1,96,500	₹ 8,995	₹ 7,440	**₹ 1,80,065**	91.6%
Wheat	₹ 1,56,000	₹ 3,240	₹ 1,440	**₹ 1,51,320**	97.0%
Maize	₹ 1,33,000	₹ 3,890	₹ 3,720	**₹ 1,25,390**	94.2%
TOTALS	**₹ 7,02,500**	**₹ 21,915**	**₹ 16,600**	**₹ 6,63,985**	**94.5%**

Figure 5. Consolidated Crop Profitability Analytical View

3.8 Systematic Testing, Verification & Test Cases

The system was evaluated against 11 test scenarios to verify that business invariants and database integrity constraints are enforced under both normal and invalid edge-case operations.

Table 15. Systematic Integrity and Validation Test Case Execution Audit

Test ID	Target Business Invariant	Test Action / Statement	Observed Engine Result & Status
TC-01	Crop dates outside season bounds	INSERT CropPlan with SowDate '2025-05-01' prior to season start	ABORT via `trg_plan_season` : Dates outside bounds. **[PASS]**
TC-02	Chronological date integrity	INSERT CropPlan with HarvestDate earlier than SowDate	CHECK violation: `CHECK(HarvestDate > SowDate)`. **[PASS]**
TC-03	Positive input quantity check	INSERT Application with negative quantity (-5 kg)	CHECKviolation: `CHECK(Qty > 0)` rejected. **[PASS]**
TC-04	Application within crop duration	INSERT Application with AppDate outside crop duration	ABORTvia `trg_app_dur` : Outside duration. **[PASS]**
TC-05	Input issue within available stock	INSERT Application requesting 99,999 kg exceeding stock	ABORTvia `trg_app_dur` : Insufficient stock. **[PASS]**
TC-06	Non-negative harvest quantity	INSERT Harvest with QtyKg = -10 kg	CHECK violation: `CHECK(QtyKg >= 0)` rejected. **[PASS]**
TC-07	Sales within available harvest stock	INSERT Sales order for 999,999 kg exceeding harvest lot	ABORT via `trg_sale_stock` : Exceeds stock. **[PASS]**
TC-08	Foreign key referential validation	INSERT Field referencing nonexistent FarmID = 99	FOREIGN KEY constraint failed; rejected. **[PASS]**
TC-09	Soil test chemical bounds	INSERT SoilTest with out-of-range pH value of 15.0	CHECK violation: `CHECK(pH BETWEEN 0 AND 14)`. **[PASS]**
TC-10	Unique crop naming constraint	INSERT Crop with duplicate CropName 'Rice'	UNIQUE constraint failed: duplicate key. **[PASS]**
TC-11	Parent-child referential integrity	DELETE FROM Farm WHERE FarmID = 1 with active fields	FOREIGN KEY restriction aborted deletion. **[PASS]**

Verification Results & Operational Audit:

All 11 test cases successfully passed verification. Invalid transactions were rejected by the database engine without data corruption. Legitimate operations executed with ACID compliance, confirming that business rules are enforced at both the UI and database levels.

CHAPTER 4: CONCLUSION, LIMITATIONS & FUTURE ENHANCEMENTS

The **FarmDB** project provides a comprehensive relational database management system for crop planning and farm input management. The system transitions agricultural record-keeping from ad-hoc spreadsheets to a normalized database decomposed to 3NF/BCNF. Key business rules—including seasonal date validation, stock-bounded pesticide issues, and harvest-capped sales allocations—are strictly enforced via triggers, CHECK constraints, and foreign keys. The companion software application supports end-to-end agricultural management, including soil nutrient tracking, irrigation logs, labor records, and crop profitability analytics.

4.1 Limitations and Future Enhancements

- **Automated Weather & IoT Telemetry Integration:** Current irrigation records are entered manually. Future iterations can integrate IoT soil-moisture sensor streams via MQTT and pull real-time weather forecasts to dynamically automate irrigation scheduling.
- **Spatial GIS Polygon Mapping:** The current system represents plot parcels using numerical acreage values. Integrating spatial extensions (PostGIS) would enable geospatial boundary mapping and satellite NDVI crop health visualization.
- **Automated Pest Outbreak Machine Learning:** The system can be extended with predictive machine learning models to forecast pest outbreaks based on historical soil test data, ambient humidity, and pesticide efficacy records.

CHAPTER 5: REFERENCES (IEEE STYLE)

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[4] SQLite Consortium, “SQLite Documentation and SQL Dialect Reference,” SQLite.org, 2024. [Online]. Available: https://www.sqlite.org/docs.html. [Accessed: Oct. 10, 2026].

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APPENDIX A: COMPLETE DATABASE DDL & ANALYTICAL SCRIPT

Below is the complete SQL DDL schema and trigger implementation script for the **FarmDB** database:

```
PRAGMA foreign_keys = ON; CREATE TABLE Farm ( FarmID INTEGER PRIMARY KEY, FarmName TEXT NOT NULL UNIQUE,
Location TEXT NOT NULL, OwnerName TEXT NOT NULL ); CREATE TABLE Field ( FieldID INTEGER PRIMARY KEY, FarmID IN TEGER NOT NULL
REFERENCES Farm(FarmID) ON DELETE RESTRICT, FieldName TEXT NOT NULL, AreaAcres REAL NOT NULL CH ECK(AreaAcres > 0), SoilType
TEXT DEFAULT 'Loam', UNIQUE(FarmID, FieldName) ); CREATE TABLE Season ( SeasonID INTEGER PRIMARY KEY, SeasonName TEXT NOT
NULL UNIQUE, StartDate TEXT NOT NULL, EndDate TEXT NOT NULL, CHECK(En dDate > StartDate) ); CREATE TABLE Crop ( CropID INTEGER
PRIMARY KEY, CropName TEXT NOT NULL UNIQUE, DurationD ays INTEGER NOT NULL CHECK(DurationDays > 0) ); CREATE TABLE Variety (
VarietyID INTEGER PRIMARY KEY, CropID I NTEGER NOT NULL REFERENCES Crop(CropID), VarietyName TEXT NOT NULL, UNIQUE(CropID,
VarietyName) ); CREATE TABL E SoilTest ( TestID INTEGER PRIMARY KEY, FieldID INTEGER NOT NULL REFERENCES Field(FieldID), TestDate
TEXT NOT NULL, pH REAL NOT NULL CHECK(pH BETWEEN 0 AND 14), N REAL NOT NULL CHECK(N >= 0), P REAL NOT NULL CHECK(P >= 0),
K REAL NOT NULL CHECK(K >= 0) ); CREATE TABLE CropPlan ( PlanID INTEGER PRIMARY KEY, FieldID INTEGER NOT N ULL REFERENCES
Field(FieldID), SeasonID INTEGER NOT NULL REFERENCES Season(SeasonID), VarietyID INTEGER NOT NU LL REFERENCES Variety(VarietyID),
SowDate TEXT NOT NULL, HarvestDate TEXT NOT NULL, Status TEXT NOT NULL DEFAU LT 'Planned' CHECK(Status
IN('Planned','Sown','Harvested')), CHECK(HarvestDate > SowDate) ); CREATE TABLE Inpu t ( InputID INTEGER PRIMARY KEY, InputName TEXT
NOT NULL UNIQUE, Type TEXT NOT NULL CHECK(Type IN('Fertilize r','Seed','Pesticide')), Unit TEXT NOT NULL, UnitCost REAL NOT NULL
CHECK(UnitCost > 0), StockQty REAL NOT NUL L DEFAULT 0 CHECK(StockQty >= 0) ); CREATE TABLE Application ( AppID INTEGER PRIMARY
KEY, PlanID INTEGER NOT N ULL REFERENCES CropPlan(PlanID), InputID INTEGER NOT NULL REFERENCES Input(InputID), AppDate TEXT NOT
NULL, Qt y REAL NOT NULL CHECK(Qty > 0) ); CREATE TABLE IrrigationEvent ( IrrID INTEGER PRIMARY KEY, PlanID INTEGER NOT NULL
REFERENCES CropPlan(PlanID), IrrDate TEXT NOT NULL, Method TEXT NOT NULL CHECK(Method IN('Flood','Dri p','Sprinkler')), WaterKL REAL
NOT NULL CHECK(WaterKL > 0) ); CREATE TABLE Labour ( LabourID INTEGER PRIMARY K EY, PlanID INTEGER NOT NULL REFERENCES
CropPlan(PlanID), WorkerName TEXT NOT NULL, Activity TEXT NOT NULL, Wor kDate TEXT NOT NULL, Hours REAL NOT NULL CHECK(Hours > 0),
WagePerHour REAL NOT NULL CHECK(WagePerHour > 0) ); CREATE TABLE Harvest ( HarvestID INTEGER PRIMARY KEY, PlanID INTEGER NOT
NULL REFERENCES CropPlan(PlanID), Har vestDate TEXT NOT NULL, QtyKg REAL NOT NULL CHECK(QtyKg >= 0) ); CREATE TABLE Buyer (
BuyerID INTEGER PRIMARY KEY, BuyerName TEXT NOT NULL UNIQUE, Phone TEXT ); CREATE TABLE Sales ( SalesID INTEGER PRIMARY KEY,
HarvestID INTEGER NOT NULL REFERENCES Harvest(HarvestID), BuyerID INTEGER NOT NULL REFERENCES Buyer(BuyerID), SaleDate T EXT
NOT NULL, QtyKg REAL NOT NULL CHECK(QtyKg >= 0), PricePerKg REAL NOT NULL CHECK(PricePerKg >= 0) );
```

APPENDIX B & C: PYTHON CONNECTOR & POPULATION AUDIT

Appendix B: Python Database Connectivity & Transaction Implementation

Representative Python backend connection script demonstrating parameterized queries and transaction control:

```
import sqlite3 class FarmDatabase: def __init__(self, db_path="farm_db.sqlite"): self.conn=sqlite3.c
onnect(db_path) self.conn.execute("PRAGMA foreign_keys = ON;") self.conn.row_factory = sqlite3.Rowdefadd_c
rop_plan(self, field_id, season_id, variety_id, sow_date, harvest_date): """Creates a crop planwithparamet
erized safety and trigger enforcement.""" cursor = self.conn.cursor() try: cursor.execute("BEGINTRANSACTIO
N;") cursor.execute(""" INSERT INTO CropPlan (FieldID, SeasonID, VarietyID, SowDate, HarvestDate,Status)VA
LUES (?, ?, ?, ?, ?, 'Planned'); """, (field_id, season_id, variety_id, sow_date, harvest_date))self.conn.c
ommit() return True, "Crop plan successfully registered." except sqlite3.IntegrityError as err:self.conn.ro
llback() return False, f"Integrity check failed: {err}" def issue_input(self, plan_id, input_id,app_date,q
ty):"""Records input application with trigger-based stock verification.""" cursor = self.conn.cursor()try:
cursor.execute("BEGIN TRANSACTION;") cursor.execute(""" INSERT INTO Application (PlanID, InputID,AppDate,Q
ty)VALUES (?, ?, ?, ?); """, (plan_id, input_id, app_date, qty)) self.conn.commit() return True,"Inputiss
uedand stock deducted." except sqlite3.IntegrityError as err: self.conn.rollback() return False,f"Stockis
suerejected: {err}"
```

Appendix C: Representative Baseline Population Audit

Table 16. Implemented Prototype Data Inventory Audit

Table Entity	Record Count	Sample Verified Data Attributes	Status
Farm & Field	2 Farms, 4 Fields	Green Valley Farm (Hyderabad), Sunrise Agro (Warangal); 15.0 Total Acres	Verified
Crop & Variety	4 Crops, 6 Varieties	Rice (BPT-5204, MTU-1010), Cotton (RCH-2), Maize (DKC-9144), Wheat (HD-2967)	Verified
SoilTest	5 Test Assays	pH range 6.5–7.9; Nitrogen 190–240 kg/ha; Phosphorous 15–25 kg/ha	Verified
CropPlan	6 Cultivation Plans	Kharif 2025 & Rabi 2025-26 operational crop plans	Verified
Input & Stock	5 Input Catalogs	Urea (2000 kg), DAP (1200 kg), Chlorpyrifos (80 L), Seed (300 kg)	Verified
Application	10 Material Issues	1,040 kg chemical inputs issued across plots	Verified
Irrigation	7 Delivery Events	1,540 kL cumulative water across Flood, Drip, Sprinkler methods	Verified
Harvest & Sales	5 Lots, 6 Sales	32,900 kg total yield; ₹7,02,500 total revenue generated	Verified